

LABORATORY RECORD**EXP NO.: 8 FABRIC NETWORK DEPLOYMENT AND MANAGEMENT****AIM**

To deploy, manage, monitor, and maintain a Hyperledger Fabric network by creating the network, verifying network components, deploying chaincode, monitoring Docker containers, and performing network shutdown and cleanup operations.

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Verify Running Docker Containers

```
docker ps
```

Step 4: Display Docker Images

```
docker images
```

Step 5: Check Network Status

```
docker ps -a
```

Step 6: Deploy Asset Transfer Chaincode

```
./network.sh deployCC -ccl javascript -ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

Step 7: Verify Installed Chaincode

```
peer lifecycle chaincode queryinstalled
```

Step 8: Monitor Peer Logs

```
docker logs peer0.org1.example.com
```

Step 9: Monitor Orderer Logs

```
docker logs orderer.example.com
```

Step 10: Verify the Channel

```
peer channel list
```

Step 11: Stop the Fabric Network

```
./network.sh down
```

Step 12: Clean Docker Resources

```
docker system prune -f
```

PROGRAM

Fabric Network Management & Monitoring Shell Script (fabric_network_manager.sh):

```
#!/bin/bash
# Script for complete Hyperledger Fabric Network Lifecycle Management and
# Monitoring

cd ~/fabric-dev/fabric-samples/test-network

# 1. Network Startup & Channel Creation
echo "Starting Fabric Network..."
./network.sh up createChannel -ca

# 2. Container & Image Verification
docker ps
docker images

# 3. Deploy Chaincode
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/
chaincode-javascript

# 4. Verify Chaincode & Channel
peer lifecycle chaincode queryinstalled
peer channel list

# 5. Monitor Service Logs
docker logs peer0.org1.example.com | tail -n 20
docker logs orderer.example.com | tail -n 20

# 6. Teardown and Cleanup
./network.sh down
docker system prune -f

echo "Fabric Network Lifecycle Executed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting Certificate Authorities...
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Fabric Network Started Successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
```

CONTAINER ID	IMAGE	COMMAND
8a9f21b10c9d	hyperledger/fabric-peer:2.5.0	"peer node start"
3c2b11e9a4f2	hyperledger/fabric-peer:2.5.0	"peer node start"
1f9a88e22d3b	hyperledger/fabric-orderer:2.5.0	"orderer"
d4e5f6a1b2c3	hyperledger/fabric-ca:1.5.6	"sh -c 'fabric-ca-server'"
a1b2c3d4e5f6	hyperledger/fabric-ca:1.5.6	"sh -c 'fabric-ca-server'"

Up 2 minutes peer0.org1.example.com 2 minutes
Up 2 minutes peer0.org2.example.com 2 minutes
Up 2 minutes orderer.example.com 2 minutes
Up 2 minutes ca_org1 2 minutes
Up 2 minutes ca_org2 2 minutes

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
hyperledger/fabric-peer	latest	a43e098a12b3	3 months ago	158MB
hyperledger/fabric-orderer	latest	c92d110f44e1	3 months ago	98.4MB
hyperledger/fabric-ca	latest	f82e3401ab12	3 months ago	150MB
hyperledger/fabric-tools	latest	e21c9983dd5a	3 months ago	435MB

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -ccl
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode deployed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer lifecycle chaincode
queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:a7629b3a783e49321c8410d8a5f829a3, Label: basic_1.0

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs
peer0.org1.example.com | tail -n 3
Peer started successfully
Listening on port 7051
Chaincode installed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-
network$ docker logs orderer.example.com | tail -n 3
Orderer started
Channel mychannel initialized
Transactions processed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Removing containers...
Removing Docker Network...
Network stopped successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker system prune -f
Deleted Containers: 8b2c11e34f01
Deleted Networks: fabric_test
Total reclaimed space: 420.5MB
```

RESULT

Hence the experiment was done successfully.